Finite-depth affine source returns for Dirichlet L-functions

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Abstract

We describe a five-stage affine source construction that retains the actual residual reduction through recursive LOW steps and counts the returned HIGH records jointly over their complete arithmetic histories. The source weight keeps every intermediate row scale; common reduction is credited only at its native valuation-cone rate. Exact rational certificates give a positive budget for all later HIGH bins. Backward induction closes the uniform whole-root moment, and the inherited detector and weighted-numerator method gives the all-moduli Dirichlet nonvanishing boundary 0.8748087. The accompanying Lean proof constructs the moment from the actual finite sources and checks the resulting unconditional theorem.

1 The initial moment and the nonvanishing theorem

Work in the Eisenstein Hecke framework of OpenAI and Collet [1, 2]. Its global zero supremum is denoted by β. The native input price is $\kappa_0 = 4499/6000$; it is an established input, independent of the conclusion. Put

$$\vartheta = \frac{8748087}{10^7}, \qquad \lambda = 3.741331371862391\ldots, \quad B = 0.982029506308942\ldots, \quad s_+ = 0.04038098. \tag{1}$$

The exact rational λ, B are specified in the accompanying parameter file. All displayed decimal junctions and widths below are exact terminating rationals, except the two explicitly truncated values in (1).

The source construction gives the following moment. Let $Z > 1$ be the root radius, $m_i = \log_Z X_i$, and $s = \sum_j w_j$ the selected prime width. Write $\mathcal{E}_\eta(\Phi, \mathbf{p})$ for the literal initial energy: its original prime pools, annular coefficients, fixed mask, fixed Q, and seed one are those of the inherited initial-moment theorem. The radial control is $\mathcal{D}(\Phi)$ and the finite profile control is $\mathcal{S}_{\mathscr{S}}(\mathbf{p})$.

Proposition 1 (Whole-root moment)**.** *Suppose* $51/100 \le \beta \le 7/8$. *For every* $\varepsilon > 0$ *a positive prime-slot mesh is chosen before the ambient cardinality and weighted source. For each fixed original weighted source and admissible* η *unchanged by excluding the unit ideal, fix its inherited* Q *and modulus/support data. There are a finite seminorm set* $\mathscr{S}$*, a degree* D*, and* $C > 0$*, chosen before* Z *and the moving profiles, such that eventually*

$$\mathcal{E}_\eta(\Phi, \mathbf{p}) \le C\mathcal{D}(\Phi)\mathcal{S}_{\mathscr{S}}(\mathbf{p})^2(1 + |t| + T)^D Z^{1+\varepsilon}, \tag{M}$$

for all shifts bounded by T*, both reflection orientations, and all actual initial sources satisfying*

$$\frac{81}{200} \le m_i \le \frac{264}{625}, \qquad \frac{17}{500} \le s \le s_+, \qquad m_1 + m_2 + \lambda s \le B.$$

The profile supports lie in $[1/4, 9/4]$ *and the original annular exponent is* $33/50$*. The same proved estimate supplies the detector and conjugated numerator, with their original source data.*

Theorem 1 (Nonvanishing)**.** *For every* $n \ge 1$*, every Dirichlet character* χ *modulo* n*, and every* $z \in \mathbb{C}$*,*

$$\Re z > \vartheta, \qquad \neg(\chi = 1 \text{ and } z = 1) \quad \Longrightarrow \quad L(\chi, z) \neq 0.$$

ArviFoxy's two-level construction [4] returns its second-step HIGH part instead of estimating it independently. We extend that organization to one root step and four inner stages. The new issue is preserving a single actual source, its residual factor, and its raw weight across every depth. The following two sections give the corresponding payment and count.

2 The actual source, its residual factor, and its raw scale

Normalize logarithms by $\log Z$. At one physical source let C_1, C_2 be the two actual common ideals, with frozen nonzero allocation coefficients. For the selected residual ideal L and allocation a, retain

$$F = \log_Z\big(\text{formalReductionFactor}(L, a)\big) \geq 0.$$

If A is the original log-volume, z the selected width, and u the retained nominal width, the native supported-source identities give

$$\begin{aligned} n_1 + n_2 + z' &\leq A - \log_Z \mathrm{N}C_1 - \log_Z \mathrm{N}C_2 - F + P, \\ n_1 + n_2 + 6\kappa_0 z' - u &\leq G + 6a_0 + \omega + P - H_C - F. \end{aligned} \tag{2}$$

Here P is the measured support padding, G the parent's old-capacity defect, a_0 its amplifier, and H_C the actual common-norm payment after the nominal-width subtraction. Both inequalities concern the *same* L, a and the same supported source; (2) follows by subtracting the two raw allocation reductions from the original volume and then taking logarithms.

This preservation is compatible with source selection. For a finite positive source sum, $E \leq \sum_i c_i E_i$ with $c_i \geq 0$, choose an index maximizing E_i/B_i, where $B_i > 0$ is its original reciprocal/Cauchy normalization. Then

$$E \leq \Big(\sum_i c_i B_i\Big) \frac{E_{i_*}}{B_{i_*}}. \tag{3}$$

Thus (2) and (3) retain the same source and residual factor. Finite physical choices retain actual provenance. When heights vary, a factor-two supremum selects one whole licensed arithmetic majorant with its full path and count receipts; it need not be an attained analytic run. Terminal depths are not maximized separately.

Lemma 1 (Raw attachment)**.** *At stage j let U_j be the physical unit, $Z_j = Z^{U_j}$ the local base, m_j the signed first-mass dyad loss, and $X_0, \dots, X_3$ the native coordinates in the original base. Set $\widehat{X}_0 = X_0 + U_j m_j$. If the LOW edge deletes actual width D_j, its raw exponent satisfies*

$$e_j = M_j + \widehat{X}_0 + q_j D_j + e_{j+1} + \epsilon_j, \qquad M_{j+1} + \log_Z \mathrm{N}(v_{j+1}) - \log_Z \mathrm{N}(v_j) \leq X_2 + P_j. \tag{4}$$

Here v_j is the actual prefix multiplier, M_j the actual row scale, and ϵ_j contains the remaining paid finite fees. The historical missing radical X_1 enters the count with coefficient one.

Proof. The source contains $\widehat{X}_0 + a_j + q_j D_j$; the AtState factor contains $M_j - a_j$. Thus a_j cancels and M_j remains. The mass term is not assumed small. The *same* first bin supplies a side lower bound b_j and label-side exponent S_j with

$$m_j = 3\delta_j - b_j, \qquad m_j + S_j \leq 3\delta_j + 6\log_{Z_j} 2.$$

This is the native shifted-coordinate payment: the large mass loss cancels its matching label side before iteration. The actual multiplier identity gives the relative-scale inequality in (4). Substitute it and $L_{j+1} \leq X_3 - D_j - e_{\mathrm{vol}} + P'_j$ in the child payment to obtain

$$(e_j - M_j) + (X_1 + \mathrm{miss}_{j+1}) + \rho(q)\big(r_{j+1} - \log_Z \mathrm{N}v_j\big).$$

There is no reciprocal cancellation of the unnormalized recursive LOW term. The volume credit is qe_{vol}; common reduction already in X_3 keeps only its native valuation-cone credit. □

A volume majorant requires compensation. For native local volume A, a lift by Δ changes (A, m, X_0, X_3) to $(A + \Delta, m - \Delta, X_0 + U\Delta, X_3 + U\Delta)$. Hence $\widehat{X}_0$, and therefore the complete raw edge weight, is unchanged; the label side increases by Δ and its cancellation also survives. The lifted m is a compensated coordinate, not a new mass estimate. At the root $U_0 = 1$ and $\Delta = A_0 - \widetilde{A}_0 \geq 0$, where $\widetilde{A}_0$ is the actual volume and A_0 its root-cell majorant. The volume coordinates below use this compensated lift, not an unpriced replacement of the actual volume.

A narrow, exhausted, old-capacity or fully reflected child returns an empty LOW-continuation forest using the native direct/old bound. A continuing child uses the next actual region. One genuine small/coarse height splice preserves the return linearly. Root radius Z, seed one and row scale one follow from the initial source reduction; the fixed conductor's logarithmic fit loss is absorbed after its eventual threshold.

3 Five-stage payment and a single terminal count

For $q \le 3/4$, put

$$\rho(q) = 1 - \frac{10q}{3(5-2q)}, \quad c(q) = 5 - 6\rho(q), \quad h(q) = \max\{3/19,\, 4\rho(q) - 5/4\}.$$

The five junctions are

$$(q_0, q_1, q_2, q_3, q_4) = (.638021915, .676506091, .693781289, .711716695, .730301837). \tag{5}$$

Starting with $(k_4, I_4) = (\kappa_0, 1)$, the preceding output region is defined recursively, for $j = 3, 2, 1, 0$, by

$$K = \frac{6(k_{j+1} - q_{j+1}I_{j+1})}{q_{j+1}c(q_{j+1})}, \quad 6k_j - 1 = \frac{6k_{j+1} - 1}{1 + Kq_{j+1}}, \quad I_j = \frac{I_{j+1} + K(1 - \rho(q_{j+1})) - 10^{-8}}{1 + Kq_{j+1}}. \tag{6}$$

The inner affine slopes are $6k_{j-1}$. Their first-error channels use prime fraction $99/100$; the terminal step uses the original price κ_0.

For an actual path, let U_j, z_j be its retained and slot widths, and A_j its compensated volume coordinate, all in the original base. Write $V_j \ge 0$ for missing-width loss, $R_j \ge 0$ for the augmented common/residual reduction and $\ell_j \ge 0$ for additional slot loss. The source gives, up to explicitly paid rounding,

$$U_{j+1} = U_j - 6a_j - V_j, \quad A_{j+1} = A_j - d_j - V_j - R_j, \quad z_{j+1} = z_j - d_j - \ell_j, \qquad \ell_j \le V_j + R_j. \tag{7}$$

Here d_j is the model deletion and a_j the affine amplifier; the actual detector deletion and finite mesh surcharge remain in (4). Let $V = \sum_{i<j} V_i$, $R = \sum_{i<j} R_i$, $a = \sum_{i \le j} a_i$, $P = \sum_{i<j} q_i d_i$, $W = 1 - V - 6a$ and $L = A_0 - V - R - \sum_{i<j} d_i$, where A_0 is the uniform root-cell upper bound for the original volume. The terminal payment licensed by the actual source is

$$J_{\mathrm{paid}}(q) = (1 - \rho(q))W - qL + a - P + \big(1 - \rho(q) - h(q)\big)V - h(q)R. \tag{8}$$

The $-h(q)R$ correction prevents counting common reduction twice. For $q \ge q_j$, the uncorrected first four terms have increment

$$\left(\frac{(50/3)W}{(5-2q)(5-2q_j)} - L\right)(q - q_j). \tag{9}$$

Here $W \ge 0$. Both correction coefficients in (8) are nondecreasing in q; a junction certificate and $\frac{50W}{3(5-2q_j)^2} - L \ge 0$ pay every later bin. The exact certificates give $J_{\mathrm{paid}}(q) > 10^{-7}$ for each returned inner depth, including the paid coordinate errors; the root own-HIGH part uses its separate marked estimate. Each affine certificate is an identity $f = \gamma + \sum_i t_i g_i$, $t_i \ge 0$: when $g_i \ge -E$ it gives $f \ge \gamma - E\sum_i t_i$. This also pays the finite moving grid.

The terminal HIGH estimate is applied once, over complete histories. A key retains both common-label pairs, their inactive subsets, the native seed, the four dyads and the orientation. If $m(H) = \sum_{i<j} X_{1,i}$ is the historical missing exponent, the prefix-fibre count is $\ll_\epsilon Z^{m+\epsilon}$ at a fixed multiplier ideal. Squarefree-divisor bounds pay the seed and fixed dyad boxes pay the remaining multiplicity. The residual L, a, F remains source metadata, not a free count coordinate. Thus the native zero-density estimate gives

$$\sum_H w_H \,\mathrm{HIGH}_H \ll CK(1+T)^D \prod_{i<j}(|\mathcal{B}_i| + 1) Z^{E+\epsilon}, \tag{10}$$

provided $w_H Z^{a_\sigma + b_\sigma + \rho(q_\sigma) r_\sigma + m(H)} \le Z^E$ for every native class σ. Here $a_\sigma, b_\sigma, r_\sigma$ are its original weighted, missing and row exponents and $q_\sigma = 2\beta_\sigma - 1$. Equations (4), (8) supply this payment with the original w_H. The zero-height exponent can be chosen before the box budget, class cap and fixed analytic return coefficient; these affect only the eventual threshold.

4 Closing the moment and applying the detector

The source construction is finite. At the last inner stage the native LOW estimate is terminal. At each preceding stage the actual LOW source produces one whole licensed returned forest, while the local HIGH source produces its own native record. The old allowance is summed with its original first-seed reciprocal; the returned forest is not divided by that seed norm. Residual projector shells, both ray sums and positive buffered bins keep the original normalization and introduce only their fixed constants and subpower losses. The three-height splice is paid once. Finite unions of seminorms and maxima of degrees precede every moving source.

Proof of Proposition 1. The terminal source starts backward induction through the four inner stages. Each preceding step invokes the actual canonical, OLD, coarse and physical sources on the same node; the resulting complete receipt supplies the next step. All primitive constants and thresholds are chosen over the finite original subsets and moving ledger grid before Z, uniformly under rebasing. The root LOW source then invokes this proved induction and the joint count; its marked HIGH estimate supplies the separate root record. The initial source reduction and a final coarse-height extension give (M). Thus the literal root-moment interface is constructed from the source proof. □

Proof of Theorem 1. Suppose $\beta > \vartheta$. The inherited unconditional bound $\beta \le 7/8$ and $\vartheta > 51/100$ admit Proposition 1. Apply it to the actual detector dictionary and conjugated weighted-numerator source. Both consumers prove the required excluded-unit character identity. Finite dictionary unions give common seminorms, degrees and thresholds before Z. Keep the outer detector density δ distinct from the inner HIGH q.

For inverse length r, plain length m, and amplitude p, the inverse, affine-plain and long-witness counts have exponents

$$1-\delta\big(p+(1-p)r\big),\quad 1-2\delta m-2\delta p\frac{B-2m}{\lambda},\quad 1-\delta+(5/6-\delta)(r+m-1). \tag{11}$$

Use the affine count only when its actual widths lie in (M)'s window; the capped-width and plain-top branches cover the adjacent ranges, and the inherited count covers the complement. The inverse exponent $5/6$ is unchanged. The affine balance, for example, is

$$R_{\mathrm{aff}}=1-\delta+\frac{(5/6-\delta)\delta(1-p)(1-2pB/\lambda)}{(5/6-\delta)(3-p-4p/\lambda)+\delta(1-p)(2-4p/\lambda)}.$$

Set $h=0.79716116522$, $\ell=11/3-4\vartheta$, $y=h-2\ell$ and $\kappa_*=2\vartheta-1$. The old and affine capacities are

$$C_O(d)=\frac{\min(d,2y-d)}{6\kappa_*},\qquad C_N(d)=\frac{2y-(2-B)d}{\lambda}.$$

The inherited weighted comparison, with its selected count exponent R, is

$$\frac{1+\delta}{2}+\frac{13h}{75}-\frac{y}{2}-\frac{\ell}{2}+\ell\delta p+d\left(\frac14-\frac{17}{50}\right)-\frac{\delta pC}{2}+\frac{3dR}{4}\ \le\ \vartheta-\frac{1}{50000000}. \tag{12}$$

The rational endpoint certificates cover $0\le\delta\le3/4$, $0\le p\le1/2$, with positive denominators and all source admissions. They use the old-capacity ends and the affine-capacity junction and top; affinity in d supplies the intervals. Clearing denominators leaves polynomials expressed in the nonnegative tensor Bernstein basis, so their positive coefficients prove (12) on each closed rectangle.

The standard feedback comparison extends (12) from κ_* to the actual $2\beta-1$. Choose the remaining positive losses inside its strict reserve before the target characters and heights. Levy's weighted numerator and Collet's continuation theorem [3, 2] then improve the same global zero supremum, a contradiction. Hence $\beta\le\vartheta$. The inherited Hecke-to-Dirichlet transfer gives the conclusion, with the principal pole excluded. □

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References

[1] OpenAI, *Quasi-Riemann hypothesis formalization*, framework and manuscript, snapshot `fd4aeeb2`.
[2] Hailey Collet, affine moment and continuation theory, PR42.
[3] Akash Levy, weighted numerator, PR6.
[4] ArviFoxy, *A second affine step for the LOW children*, PR67.
[5] timbalud, asymmetric marked counts, PR50; DaysSky, affine detector, PR56.
[6] Zhongpai Gao, balanced common geometry and adaptive marked source, PR63.